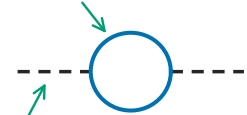


$\psi_2 = \bar{\psi}_1$ here

vanishes by $p \rightarrow -p$

can cut off in other ways, but is finite anyway



$$= 6g \int_{-\pi/a}^{\pi/a} \frac{dp}{2\pi} \frac{-ip + m}{p^2 + m^2} = \frac{6g}{\pi} \arctan\left(\frac{\pi}{2ma}\right) \sim 3g + \mathcal{O}(ma)$$



$$= \frac{6g}{L} \sum_{k=0}^{L-1} \frac{-2i \sin(\pi k/L) e^{i\pi k/L} + m_W a}{\sin^2(\pi k/L) + m_W^2 a^2} \xrightarrow{a \rightarrow 0} 6g$$

where $m_W a = ma + 2\sin^2\left(\frac{\pi k}{2L}\right)$ due to Wilson term